C. V. RAMAN GLOBAL UNIVERSITY

ODISHA

A REPORT ON

POWER SYSTEMS

CLASSIFICATION OF DISTRIBUTION SYSTEMS

AND

INFLUENCE OF WORKING VOLTAGE ON SIZE OF FEEDERS AND DISTRIBUTORS



GROUP 3

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

C. V. RAMAN GLOBAL UNIVERSITY
ODISHA

ABSTRACT

Electrical power distribution is an essential part of a power system that delivers electrical energy from distribution substations to consumers. The selection of a suitable distribution system and its operating voltage plays an important role in ensuring reliable, efficient, and economical power delivery.

This report presents the classification of electrical distribution systems based on the type of current, construction, and connection schemes, including radial, ring main, and interconnected systems. It also examines the influence of working voltage on the size of feeders and distributors. The relationship between voltage, current, conductor cross-sectional area, power losses, and voltage drop is discussed using fundamental electrical engineering principles and mathematical expressions.

The study highlights how an appropriate choice of working voltage can help reduce current, minimize power losses, and improve the economic performance of distribution networks. It also explains the practical factors that must be considered when selecting conductor sizes for feeders and distributors.

The report aims to develop a clear understanding of distribution system configurations and the importance of voltage selection in the design and operation of modern electrical power distribution networks.

TEAM INTRODUCTION

We are Group 3 from the Department of Electrical and Electronics Engineering at C. V. Raman Global University, Odisha. As part of our Power Systems coursework, we have undertaken a study on the topic “Classification of Distribution Systems and Influence of Working Voltage on the Size of Feeders and Distributors.”

Our group aims to understand the fundamental principles of electrical power distribution, examine the characteristics of different distribution system configurations, and explore how working voltage affects conductor sizing, power losses, and voltage regulation.

Through this study, we seek to connect theoretical concepts with practical engineering applications and develop a better understanding of the design requirements of efficient and reliable electrical distribution networks.

This report represents our collective effort to study the topic, organize technical information, and present the concepts in a clear and systematic manner.

Sl.No	Registration Number	Name
01	2401020979	Aditya Kumar Barik
02	2401021025	Abhay Kumar
03	2401021032	Jatin kumar Mallick
04	2401021047	Santosh Kumar Baliarsingh
05	2401021061	Chandramouli Shil
06	2401021062	Rounak Prajapati
07	2401021063	Aditya Prakash Mohapatra
08	2401021071	Suryakant Thakur
09	2401021080	Sanat Soyansu Jena
10	2401021106	Kuldeep Keshari
11	2401020946	Priyanka Pradhan

Guided by: Smrutirekha Samal Ma’am

4. INTRODUCTION TO ELECTRICAL DISTRIBUTION SYSTEMS

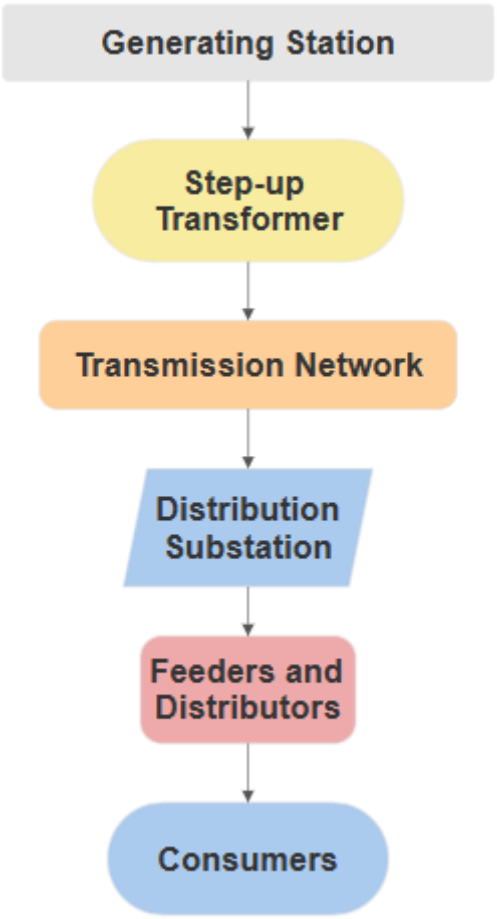
An electrical power system consists of three main stages: generation, transmission, and distribution. Electrical energy is generated at power stations, transmitted over long distances at high voltages, and then supplied to consumers through distribution networks.

An electrical distribution system is the part of the power system responsible for delivering electrical energy from distribution substations to domestic, commercial, industrial, and other consumers. It includes distribution feeders, distributors, service mains, transformers, and associated protective equipment.

The main objective of a distribution system is to provide a reliable, safe, and economical supply of electrical energy while maintaining voltage within permissible limits.

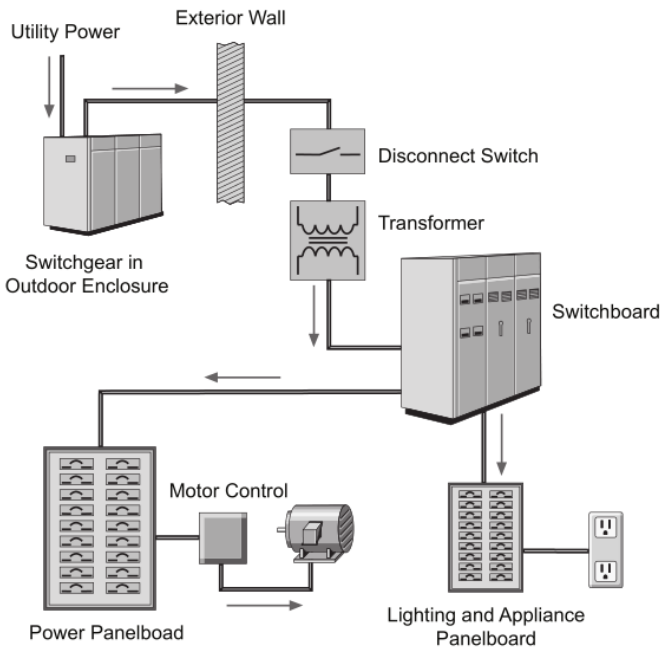
Distribution systems can be classified according to the type of current used, the number of phases and wires, the method of construction, and the network connection arrangement. Common network arrangements include radial, ring main, and interconnected systems.

The selection of a suitable distribution arrangement and working voltage is important because it affects conductor size, power losses, voltage regulation, reliability, and the overall cost of the network. A proper understanding of these factors helps engineers design efficient distribution systems capable of meeting the electrical demands of consumers.



BASIC COMPONENTS OF AN ELECTRICAL DISTRIBUTION SYSTEM

An electrical distribution system consists of several components that work together to deliver electrical energy from a distribution substation to consumers. The three main components are feeders, distributors, and service mains.



a. Feeders

A feeder is a conductor that carries electrical power from a distribution substation to distribution points or areas where loads are supplied. Normally, no consumer connections are tapped directly from the feeder along its route.

The cross-sectional area of a feeder is primarily determined by its current-carrying capacity, permissible voltage drop, thermal limits, and economic considerations.

b. Distributors

A distributor is a conductor from which electrical connections are tapped to supply individual consumers. The current flowing through a distributor varies along its length because loads are connected at different points.

The design of a distributor is mainly concerned with maintaining the voltage at consumer terminals within permissible limits. Therefore, the magnitude and distribution of loads, conductor resistance, and voltage drop are important design factors.

c. Service Mains

Service mains are the conductors that connect a distributor to an individual consumer's premises. They provide the final electrical connection through which power is supplied to domestic, commercial, or industrial installations.

Their size depends on the consumer's load, current-carrying capacity, permissible voltage drop, and applicable electrical safety requirements.

d. Importance of These Components

Feeders, distributors, and service mains perform different functions in the distribution network. Their proper design ensures safe power delivery, acceptable voltage levels, reduced electrical losses, and reliable operation.

Understanding these components is essential for studying how working voltage influences conductor size, particularly in feeders and distributors.

CLASSIFICATION OF DISTRIBUTION SYSTEMS

Electrical distribution systems can be classified according to the type of current, number of phases and wires, method of construction, and network connection arrangement. The selection of a suitable distribution system depends on the nature of the load, geographical conditions, reliability requirements, and economic considerations.

1. Classification Based on Type of Current

1i. Direct Current (DC) Distribution System

In a DC distribution system, electrical energy is distributed using direct current, which flows in one direction. DC distribution is used in applications such as battery-based systems, electric vehicle charging installations, data centres, and certain specialised industrial systems.

Advantages:

- Suitable for battery storage and DC loads.
- Avoids reactive power in purely DC circuits.
- Useful in renewable energy and modern electronic applications.

1ii. Alternating Current (AC) Distribution System

In an AC distribution system, electrical current periodically changes its magnitude and direction. AC distribution is widely used because transformers allow voltage to be increased or decreased conveniently, making it suitable for conventional power distribution.

AC distribution systems may be single-phase or three-phase.

Advantages:

- Voltage can be changed using transformers.
- Three-phase systems are suitable for industrial loads and motors.
- Widely established infrastructure makes AC distribution practical and economical.

2. Classification Based on Number of Phases and Wires

2i. Single-Phase Distribution

Single-phase distribution is commonly used for domestic lighting, fans, and other small electrical appliances. In India, typical domestic supply is approximately 230 V at 50 Hz.

2ii. Three-Phase Distribution

Three-phase distribution is commonly used in industrial installations, commercial buildings, and large electrical loads. In a typical Indian low-voltage system, the nominal voltage is approximately 415 V between phases and 230 V between phase and neutral.

Three-phase systems can transmit more power efficiently and provide a suitable supply for three-phase induction motors and other industrial equipment.

3. Classification Based on Method of Construction

3i. Overhead Distribution System

In an overhead distribution system, conductors are installed above ground using poles, towers, and suitable insulators.

Advantages:

- Lower initial installation cost.
- Faults can often be located and repaired more easily.
- Extensions and modifications are relatively simple.

Disadvantages:

- Exposed to storms, lightning, and falling trees.
- Requires clearance and suitable supporting structures.
- May affect the appearance of urban areas.

3ii. Underground Distribution System

In an underground distribution system, insulated cables are installed below ground in trenches, ducts, or conduits.

Advantages:

- Better protection from weather-related disturbances.
- Improved appearance of urban and residential areas.
- Reduced exposure to accidental contact with overhead conductors.

Disadvantages:

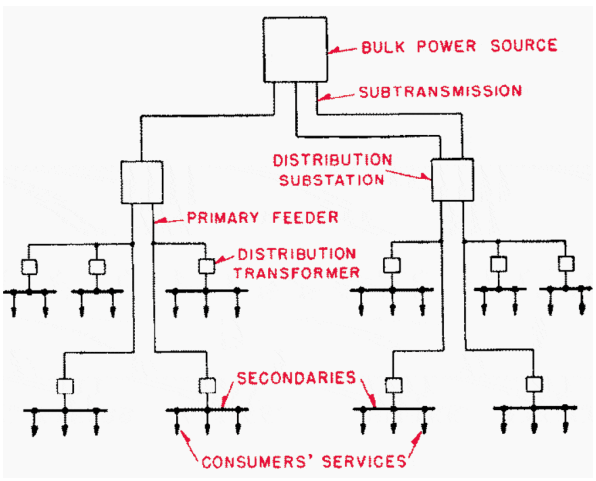
- Higher installation cost.
- Fault location and repairs can be more difficult.
- Cable installation and maintenance may require specialised equipment.

4. Classification Based on Network Connection Scheme

4i. Radial Distribution System

In a radial system, each consumer or distribution area receives power through a single main path from the supply point.

Advantages: Simple design, low initial cost, and easy operation.



Disadvantages: A fault in the supply path may interrupt power to all downstream consumers, and voltage regulation may be poorer on long, heavily loaded lines.

Applications: Rural areas, small towns, and low-cost distribution networks.

4ii. Ring Main Distribution System

In a ring main system, the distributor forms a closed loop around the load area. Depending on the network arrangement and switching equipment, consumers may receive power from either direction.

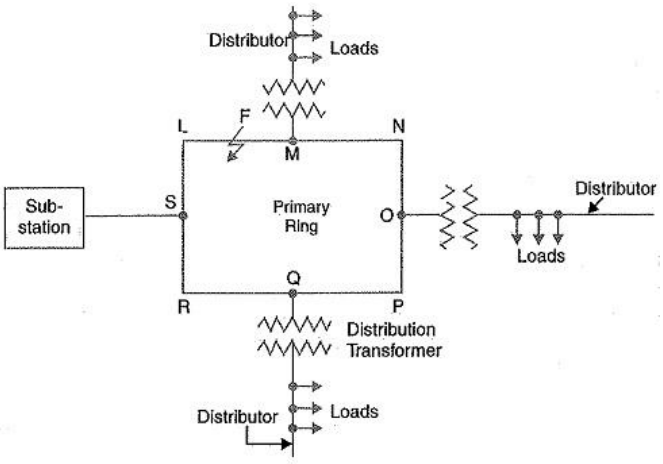


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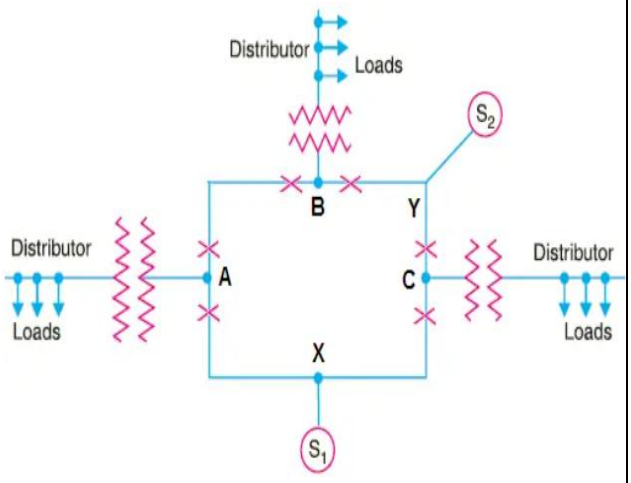
Advantages: Improved supply continuity, greater flexibility during maintenance, and potentially better voltage regulation.

Disadvantages: Higher installation cost and more complex protection and switching requirements than a simple radial system.

Applications: Urban areas, commercial complexes, and industrial installations.

4iii. Interconnected Distribution System

An interconnected system links two or more substations or distribution networks, allowing power to be supplied through multiple interconnected paths.



Advantages: High reliability, improved load sharing, and greater operational flexibility.

Disadvantages: Higher capital cost and more complex system protection, coordination, and control.

Applications: Large cities, major industrial areas, and important load centres.

5. Summary

Distribution systems differ in their electrical characteristics, construction methods, and network arrangements. Radial systems are simple and economical, ring main systems provide additional supply flexibility, and interconnected systems can provide greater reliability. The most suitable arrangement depends on load requirements, cost, operating conditions, and the desired continuity of supply.

COMPARISON OF DISTRIBUTION SYSTEMS

The performance of an electrical distribution system depends on its network arrangement, installation cost, reliability, voltage regulation, and operational complexity. The three commonly used arrangements are radial, ring main, and interconnected distribution systems.

a. Comparative Analysis

Parameter	Radial System	Ring Main System	Interconnected System
Network arrangement	Single supply path	Closed-loop path	Multiple interconnected paths
Initial cost	Low	Moderate to high	High
Reliability	Comparatively low	High	Very high
Voltage regulation	Generally poorer on long, heavily loaded lines	Generally better	Generally better
Fault isolation	Simple, but downstream loads may lose supply	Alternative supply may be available	Flexible supply and fault isolation
Protection	Relatively simple	More complex	Most complex
Operation	Simple	Moderately complex	Complex
Typical applications	Rural areas and small load centres	Urban and commercial areas	Large cities and major industrial networks

Note: The comparison is general. Actual performance depends on network design, protection, switching arrangements, conductor sizing, and loading.

b. Selection of a Suitable Distribution System

The selection of a distribution arrangement depends on the required reliability, nature of the load, installation cost, geographical conditions, and future expansion requirements.

A radial system is suitable where simplicity and low initial cost are important. A ring main system is preferred where alternative supply paths improve continuity of service. An interconnected system is useful where high reliability, load sharing, and operational flexibility are required.

c. Importance of Distribution System Selection

Choosing an appropriate distribution system helps maintain acceptable voltage levels, reduce the duration and extent of supply interruptions, and achieve economical operation. The selected arrangement must meet present demand while allowing for future load growth and safe system operation.

WORKING VOLTAGE IN ELECTRICAL DISTRIBUTION SYSTEMS

a. Definition of Working Voltage

Working voltage is the voltage at which an electrical distribution system or a particular part of the system operates under normal conditions. It is an important design parameter because it influences current, conductor size, power losses, voltage drop, insulation requirements, and the overall cost of a distribution network.

In electrical power distribution, voltage is selected according to the power demand, distance of transmission within the distribution network, type of load, safety requirements, and economic considerations.

b. Importance of Working Voltage

The choice of working voltage plays a significant role in the efficient and economical distribution of electrical energy. For a given power demand, increasing the voltage reduces the current required to deliver that power.

A reduction in current can provide several benefits:

1. **Reduced conductor size:** A lower current may allow a smaller conductor cross-sectional area, subject to thermal, voltage-drop, and mechanical constraints.
2. **Reduced power losses:** Lower current reduces resistive losses in conductors.
3. **Improved voltage regulation:** Reduced current generally lowers the voltage drop along a distribution line of given resistance and reactance.
4. **Improved economic performance:** Lower losses and potentially smaller conductors can reduce operating and material costs.
5. **Insulation requirements:** Higher working voltages require greater insulation capability and appropriate clearances, which may increase equipment costs.

c. Relationship Between Voltage, Current, and Power

For a single-phase AC system:

The active power is given by:

$P = V \times I \times \cos \varphi$

Therefore, the current is:

$I = P / (V \times \cos \varphi)$

For a balanced three-phase AC system:

The active power is given by:

$P = \sqrt{3} \times V_l \times I_l \times \cos \varphi$

Therefore, the line current is:

$I_l = P / (\sqrt{3} \times V_l \times \cos \varphi)$

Where**:**

- P = Active power in watts (W)
- V = Single-phase supply voltage in volts (V)
- V_l = Three-phase line voltage in volts (V)
- I = Single-phase current in amperes (A)
- I_l = Three-phase line current in amperes (A)
- cos φ = Power factor
- φ = Phase angle between voltage and current

For constant power and power factor, current is inversely proportional to voltage. Thus, increasing the working voltage reduces the current required to deliver the same amount of power.

d. Common Voltage Levels in Distribution Systems

Distribution networks operate at different voltage levels depending on the stage of supply and the nature of the load.

In India, common examples include:

- **11 kV:** Commonly used for primary distribution feeders.
- **415 V:** Typical three-phase, low-voltage distribution supply.
- **230 V:** Typical single-phase supply for domestic consumers.

These are common nominal values; actual system arrangements and voltage levels may vary depending on the utility and application.

e. Working Voltage and Distribution System Design

The working voltage must be selected by considering both electrical performance and practical requirements. Higher voltage can reduce current, conductor requirements under certain design conditions, power losses, and voltage drop. However, it also increases insulation requirements and may require more expensive switchgear, transformers, and protective equipment.

Therefore, the most suitable working voltage is selected by balancing the benefits of reduced current and losses against the additional cost and technical requirements of higher-voltage equipment.

INFLUENCE OF WORKING VOLTAGE ON FEEDER AND DISTRIBUTOR SIZE

a. Effect on Conductor Size

Working voltage significantly affects the current flowing through feeders and distributors. For a given power demand, increasing the voltage reduces the current required to deliver the same power.

For a balanced three-phase AC system:

$P = \sqrt{3} \times V_l \times I_l \times \cos \varphi$

Therefore:

$I_l = P / (\sqrt{3} \times V_l \times \cos \varphi)$

Where P is active power, V_l is line voltage, I_l is line current, and $\cos \varphi$ is the power factor.

A reduction in current may permit a smaller conductor cross-sectional area, provided that current-carrying capacity, permissible voltage drop, thermal limits, and mechanical strength requirements are satisfied.

b. Effect on Feeders and Distributors

Feeders: Feeders carry power from distribution substations to distribution points. Their sizes are primarily determined by current-carrying capacity and permissible voltage drop. Higher working voltage reduces current for the same power demand and may reduce the required conductor size.

Distributors: Distributors supply consumers through connections tapped along their length. Their sizes are mainly influenced by permissible voltage drop and the distribution of loads. Higher working voltage can reduce current and voltage drop, helping maintain an acceptable voltage at consumer terminals.

c. Effect on Power Losses and Voltage Drop

The resistive power loss in a balanced three-phase system is:

$P_loss = 3 \times I_l^2 \times R$

For a simplified balanced three-phase circuit with negligible reactance, the approximate line-to-line voltage drop is:

Voltage drop $\approx \sqrt{3} \times I_l \times R \times \cos \varphi$

For the same transmitted power and unchanged conductor resistance, increasing the working voltage reduces current, which in turn reduces resistive losses and generally reduces voltage drop.

d. Numerical Example

Consider a balanced three-phase system transmitting 100 kW at a power factor of 0.8 lagging. Compare its current at line voltages of 400 V and 11 kV.

Using:

$I_l = P / (\sqrt{3} \times V_l \times \cos \varphi)$

At 400 V:

$I_l = 100{,}000 / (\sqrt{3} \times 400 \times 0.8)$

$I_l \approx 180.42$ A

At 11 kV:

$I_l = 100{,}000 / (\sqrt{3} \times 11{,}000 \times 0.8)$

$I_l \approx 6.56$ A

The voltage increases by a factor of 27.5, so the current decreases by the same factor for constant power and power factor.

Assuming a constant resistance of 0.1 Ω per phase, the three-phase resistive losses are calculated using:

$P_loss = 3 \times I_l^2 \times R$

At 400 V, the loss is approximately 9.77 kW.

At 11 kV, the loss is approximately 12.9 W.

This comparison assumes identical conductor resistance in both cases to demonstrate the relationship between voltage, current, and power loss. It does not represent a direct comparison of actual 400 V and 11 kV installations, which would use appropriately

CONCLUSION:

Electrical distribution systems are essential for supplying electrical energy safely and efficiently to consumers. Different arrangements, including radial, ring main, and interconnected systems, offer varying levels of cost, reliability, and operational flexibility.

The study demonstrates that increasing working voltage for a given power demand reduces current, which can reduce conductor requirements, resistive losses, and voltage drop under suitable design conditions. However, insulation, equipment costs, thermal limits, and permissible voltage variation must also be considered.

An appropriate selection of distribution arrangement, working voltage, and conductor size is therefore essential for achieving reliable, efficient, and economical electrical power distribution.

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4. Relevant electrical distribution notes and classroom materials provided by the Department of Electrical and Electronics Engineering, C. V. Raman Global University.
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